

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO3:** Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

CODING CHALLENGE

Duration: 2hrs

1. Predict the Reinforcement Learning (RL) framework by describing the agent–environment interaction, emphasizing the importance of states, actions, and rewards, and discuss how cumulative reward influences decision-making.
2. Estimate how decision-making problems can be modeled as a Markov Decision Process (MDP), detailing the components of action space, transition probability, reward function, and discount factor
3. Discuss the role of policy and value functions in RL, and examine how state-value and action-value functions contribute to evaluating long-term benefits of actions using the Bellman Equation
4. Justify the exploration–exploitation trade-off in RL and compare strategies such as ϵ -greedy and softmax approaches in balancing learning efficiency
5. Explain how reward shaping can accelerate learning in RL and illustrate its impact on preventing suboptimal policies

ANSWERS:

1. Reinforcement Learning Framework

Problem Statement

Develop a simple Reinforcement Learning framework to demonstrate the interaction between an agent and an environment using states, actions, rewards, and cumulative reward.

Python Code

```
import random

state = 0
goal = 5
total_reward = 0

while state != goal:
    action = random.choice([1, -1])
    state = max(0, state + action)

    reward = 10 if state == goal else -1
    total_reward += reward

    print("State:", state,
          "Action:", action,
          "Reward:", reward)

print("Cumulative Reward:", total_reward)
```

Sample Output

```
State: 1 Action: 1 Reward: -1
State: 2 Action: 1 Reward: -1
State: 3 Action: 1 Reward: -1
State: 4 Action: 1 Reward: -1
State: 5 Action: 1 Reward: 10
```

Cumulative Reward: 6

Conclusion

The agent successfully reached the goal by interacting with the environment. The cumulative reward guided the agent toward better decision-making.

2. Markov Decision Process (MDP)

Problem Statement

Develop a simple Markov Decision Process model to demonstrate state transitions, rewards, action space, and discount factor.

Python Code

```
states = ["A", "B", "C"]

transition = {
    "A": ("B", 5),
    "B": ("C", 10),
    "C": ("C", 0)
}
```

```

gamma = 0.9

state = "A"

while state != "C":
    next_state, reward = transition[state]
    print(state, "->", next_state, "Reward:", reward)
    state = next_state

print("Discount Factor:", gamma)

```

Sample Output

A -> B Reward: 5
 B -> C Reward: 10
 Discount Factor: 0.9

Conclusion

The MDP successfully modeled sequential decision-making using state transitions, rewards, and a discount factor.

3. Policy and Value Function (Bellman Equation)

Problem Statement

Develop a program to calculate the state value using the Bellman Equation and demonstrate the role of policy and value functions.

Python Code

```

reward = 10
gamma = 0.9
next_state_value = 5

current_value = reward + gamma * next_state_value

print("Reward:", reward)
print("Next State Value:", next_state_value)
print("Current State Value:", current_value)

```

Sample Output

Reward: 10
 Next State Value: 5
 Current State Value: 14.5

Conclusion

The Bellman Equation estimates the long-term value of a state by combining immediate and future rewards.

4. Exploration vs Exploitation

Problem Statement

Develop a program to compare ϵ -greedy and softmax action-selection strategies for balancing exploration and exploitation.

Python Code

```

import numpy as np
import random

values = [2, 5, 8]

```

```

epsilon = 0.2

# ε-Greedy
if random.random() < epsilon:
    action = random.randint(0, 2)
    print("Exploration")
else:
    action = np.argmax(values)
    print("Exploitation")

print("Selected Action:", action)

# Softmax probabilities
exp_values = np.exp(values)
prob = exp_values / np.sum(exp_values)

print("Softmax Probabilities:", prob)

```

Sample Output

```

Exploitation
Selected Action: 2
Softmax Probabilities:
[0.0024 0.0473 0.9503]

```

Conclusion

The ϵ -greedy strategy selects either the best or a random action, while the softmax strategy selects actions according to probability, providing smoother exploration.

5. Reward Shaping

Problem Statement

Develop a program to demonstrate reward shaping and show how intermediate rewards help the agent learn an optimal policy.

Python Code

Python

Run

```

goal = 5
position = 0
reward = 0

```

```

while position < goal:
    position += 1

    if position == goal:
        reward += 100
    else:
        reward += 10

    print("Position:", position,
          "Reward:", reward)

```

Sample Output

```

Position: 1 Reward: 10
Position: 2 Reward: 20

```

Conclusion

Reward shaping accelerates learning by providing intermediate rewards, encouraging the agent to reach the goal efficiently and reducing the chances of learning suboptimal policies.

Code of Conduct:

*I **Kandukuri.Kavya-192425283** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Kandukuri.Kavya-192425283

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

		relevant examples			
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